

Assignment - 1

①

Given stock prices

(a) simple monthly return

month price

1	100
2	108
3	103
4	115
5	110

$$R_t = \frac{P_t - P_{t-1}}{P_{t-1}}$$

$$R_1 = \frac{108-100}{100} = \frac{8}{100} = 0.08 = 8\%$$

$$R_2 = \frac{103-108}{108} = \frac{-5}{108} = -0.046 = -4.6\%$$

$$R_3 = \frac{115-103}{103} = \frac{12}{103} = 0.1165 = 11.65\%$$

$$R_4 = \frac{110-115}{115} = \frac{-5}{115} = -0.0434 = -4.34\%$$

$$R_5 = \frac{110-115}{115} = \frac{-5}{115} = -0.0434 = -4.34\%$$

(b) log return:

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right)$$

$$r_1 = \ln\left(\frac{108}{100}\right) = \ln(1.08) = 0.0769 = 7.69\%$$

$$r_2 = \ln\left(\frac{103}{108}\right) = \ln(0.9537) = -0.048 = -4.8\%$$

$$r_3 = \ln\left(\frac{115}{103}\right) = \ln(1.1165) = 0.110 = 11\%$$

$$r_4 = \ln\left(\frac{110}{115}\right) = \ln(0.9565) = -0.0444 = -4.44\%$$

(c) for small returns

$$r_t \approx R_t$$

1st month $\rightarrow$ SR = 8% ; LR = 7.7% $\rightarrow$ very small difference

2nd month $\rightarrow$ SR = -4.6% ; LR = -4.48% $\rightarrow$ very very small diff

3rd month $\rightarrow$ SR = 11.65% ; LR = 11% $\rightarrow$ somewhat larger.

4th month $\rightarrow$ SR = -4.34% ; LR = -4.44% $\rightarrow$ very small diff

* The approximation is reasonably accurate for all months because $r_t \approx R_t$.
 $\rightarrow$ however least accurate is 3rd month.

② Given

$$\left. \begin{aligned} E[R_1] &= 15\% , w_1 = 0.4 \\ E[R_2] &= 7\% , w_2 = 0.6 \end{aligned} \right\} \text{--- ①}$$

linearity of Expectation property

$$\begin{aligned} E[ax + by] &= E[ax] + E[by] \\ &= aE[x] + bE[y] \end{aligned}$$

for portfolio:

$$E[R_P] = w_1 E[R_1] + w_2 E[R_2]$$

Substituting given values from Eqn ①

$$E[R_P] = (0.4)(15) + (0.6)(7)$$

$$= 6 + 4.2$$

$$= \boxed{10.2\%}$$

$$(b) \rho_{12} = \frac{\text{Cov}(R_1, R_2)}{\sigma_1 \sigma_2} \quad \text{correlation coefficient}$$

$$\hookrightarrow E[R_P] = w_1 E[R_1] + w_2 E[R_2]$$

The Expected return formula does not contain any correlation term, so it doesn't depend.

→ correlation affects only portfolio variance / risk.

③ Two-Asset portfolio variance

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2 \text{Cov}(X, Y)$$

→ we have to prove

$$\sigma_P^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \rho_{12} \sigma_1 \sigma_2$$

Proof:- $R_P = w_1 R_1 + w_2 R_2$

Portfolio -
 $R_P \rightarrow$ return

$$\text{Var}(R_P) = \text{Var}(w_1 R_1 + w_2 R_2)$$

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$$\text{Var}(aX + bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y) + 2ab \text{Cov}(X, Y)$$

So,

$$\text{Var}(R_p) = w_1^2 \text{Var}(R_1) + w_2^2 \text{Var}(R_2) + 2w_1 w_2 \text{Cov}(R_1, R_2)$$

$$\text{Var}(R_p) = \sigma_p^2$$

$$\text{Cov}(R_1, R_2) = \rho_{12} \sigma_1 \sigma_2$$

$$\text{Var}(R_i) = \sigma_i^2, i=1, 2$$

So, finally

$$\sigma_p^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \rho_{12} \sigma_1 \sigma_2$$

$$\rho \rightarrow -1$$

↳ Possible to eliminate risk completely

$$\sigma_p^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \rho_{12} \sigma_1 \sigma_2$$

$\rho < 1 \rightarrow$ diversification reduces risk.

$w_1^2 \sigma_1^2 =$ Asset - first Risk, $w_2^2 \sigma_2^2 =$ Second Asset Risk

$2w_1 w_2 \rho_{12} \sigma_1 \sigma_2 =$ diversification term

$\rho = 1 \rightarrow$ no diversification benefit.

(4) Given

$$\mu_1 = 18\%, \mu_2 = 8\%$$

$$\sigma_1 = 25\%, \sigma_2 = 12\%$$

$$w_1 = 0.5, w_2 = 0.5$$

$$(a) \sigma_p^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \rho_{12} \sigma_1 \sigma_2$$

different cases

$$\sigma_p^2 = (0.5)^2 (25)^2 + (0.5)^2 (12)^2 + 2 \times 0.5 \times 0.5 \times (-1) \times 25 \times 12$$

$$= (0.25)(625) + (0.25)(144) + 0.5 \times 2 \times 300$$

$$\sigma_p^2 = 156.25 + 36 + 150 \times$$

$$\lambda = \rho = -1; \sigma_p^2 = 156.25 + 36 + 150 = 342.25$$

$$\sigma_p = \sqrt{342.25} = 18.5\%$$

$$\rho = 0$$

$$\sigma_p^2 = 156.25 + 36 + 0$$

$$\sigma_p = \sqrt{192.25}$$

$$\sigma_p \approx 13.86\%$$

$$\rho = -1$$

$$\sigma_p^2 = 156.25 + 36 - 150$$

$$\sigma_p = \sqrt{42.25}$$

$$= 6.5\%$$

→ Idea here is as correlation ↓es, Portfolio risk decreases

(b) we know

Portfolio variance formula

$$\sigma_p^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \rho \sigma_1 \sigma_2$$

if $\rho = +1$

$$\sigma_p^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \sigma_1 \sigma_2$$

$$\sigma_p = (w_1 \sigma_1 + w_2 \sigma_2)^2$$

$$\sigma_p = w_1 \sigma_1 + w_2 \sigma_2$$

~~the~~ if $\rho = +1 \Rightarrow$ portfolio risk equals to weighted average of individual risks.

(c) for $\rho = -1$, $w_1^* = ?$ s.t. $\sigma_p = 0$

→ for perfect -ve correlation

$$w_1 \sigma_1 = w_2 \sigma_2 \quad \& \text{ also } w_1 + w_2 = 1$$

$$\text{So, } w_1 (25) = (1 - w_1) (12)$$

$$w_2 = 1 - w_1$$

$$25w_1 = 12 - 12w_1$$

$$37w_1 = 12$$

$$w_1 = 12/37$$

$$= 0.324$$

$$w_1^* = 0.324$$

$$w_2^* = 1 - 0.324$$

$$= 0.676$$

⑤ when two assets have correlation $\rho < 1$, their prices do not move exactly together. Becoz of this in one asset can be balanced by loss or gain in other asset which reduces overall portfolio risk.

from the portfolio variance formula

$$\sigma_p^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1w_2\rho\sigma_1\sigma_2$$

when $\rho < 1 \rightarrow$ total

that means portfolio risk becomes lower than

- weighted avg of individual risks

$\Rightarrow$ This phenomenon is called diversification effect.

If $\rho = 1$, both assets move perfectly together, so diversification does not reduce risk.

⑥ Given

Asset	w_i	σ_i
1	15%	25%
2	8%	12%
3	5%	4%

Correlations

$$\rho_{12} = 0.4$$

$$\rho_{13} = 0.1$$

$$\rho_{23} = 0.2$$

also

$$\rho_{ii} = 1$$

~~we~~ we know that

$$\text{Cov}(i,j) = \Sigma_{ij} = \rho_{ij} \sigma_i \sigma_j$$

$\downarrow$

⑦ covariance general formula

$$\Sigma_{ii} = (\sigma_i)^2 \quad \text{Cov}(1,2) = \Sigma_{12} = (0.4)(25)(12)$$

$$\Sigma_{11} = (25)^2 \quad \hookrightarrow \boxed{= 120}$$

$$\Sigma_{22} = (12)^2 \quad \text{Cov}(1,3) = \Sigma_{13} = (0.1)(25)(4)$$

$$\Sigma_{33} = (4)^2 \quad \hookrightarrow \boxed{= 10}$$

$$\text{Cov}(2,3) = \Sigma_{2,3} = (0.2)(12)(4)$$

$$\hookrightarrow \boxed{= 96}$$

W.K.T
Covariance matrix is symmetric
Then $\sigma_{12} = \sigma_{21} \dots$ so, on.

Then $\Sigma_{21} = 120$
 $\Sigma_{31} = 10$
 $\Sigma_{32} = 9.6$ $\rightarrow \Sigma = \begin{bmatrix} 625 & 120 & 10 \\ 120 & 144 & 9.6 \\ 10 & 9.6 & 16 \end{bmatrix}$

(b) A matrix is symmetric if

$$A^T = A$$

$$(\Sigma)^T = \begin{bmatrix} 625 & 120 & 10 \\ 120 & 144 & 9.6 \\ 10 & 9.6 & 16 \end{bmatrix}^T = \begin{bmatrix} 625 & 120 & 10 \\ 120 & 144 & 9.6 \\ 10 & 9.6 & 16 \end{bmatrix}$$

$$(\Sigma)^T = \Sigma \rightarrow \text{Hence proved}$$

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Given $R_f = 4\%$

$$\text{Sharpe Ratio} = \frac{E[R_M] - R_f}{\sigma_M}$$

$R_f = \text{risk free return or guaranteed return}$

Portfolio X
 $E[R_M] = 14\%$
 $\sigma_M = 22\%$

$$S_X = \frac{14 - 4}{22} = 10/22$$

$$S_X \approx 0.455$$

Portfolio for Y
 $E[R_M] = 9\%$
 $\sigma_M = 10\%$

$$S_Y = \frac{9 - 4}{10} = 5/10 = 1/2$$

$$S_Y = 0.5$$

Portfolio Z
 $E[R_M] = 20\%$

$$\sigma_P = 35\%$$

$$S_Z = \frac{20 - 4}{35} = 16/35$$

$$S_Z \approx 0.457$$

(b) higher sharpe ratio means $\Rightarrow$ better return per unit risk

$$Y > Z > X$$

Rational investor would prefer: -

Portfolio Y

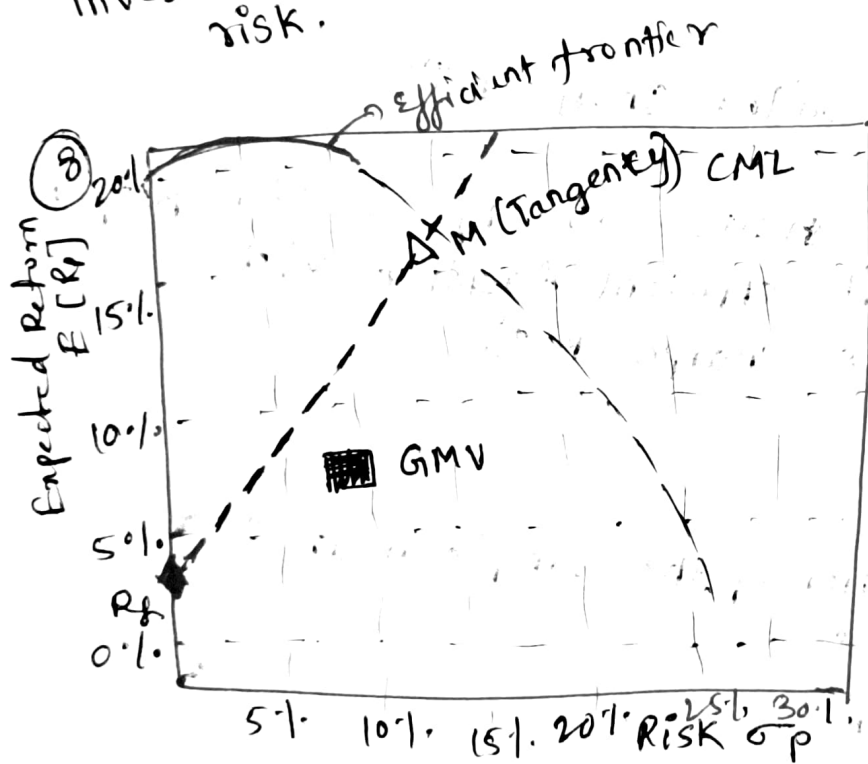
Because it has $\rightarrow$ highest sharpe ratio, best risk-adjusted return.

(c)

Portfolio Z has highest absolute return, but investor still choose Y because

Y $\rightarrow$ has lower return, but much lower risk.

$\rightarrow$ we & investors choose ^{more} about return relative to risk.



(a) GMV portfolio: the leftmost point.
The portfolio with lowest possible risk across all weight combinations.
So we can say;
GMV $\rightarrow$ minimum risk portfolio

(b) Lower half is inefficient from our Lecture notes & observing the graph above for every point on the lower half, $\exists$ a portfolio on the upper half with identical risk & higher return.

also, lower half is always dominated by corresponding point on upper half.

• lower half = inefficient

that's why rational investors choose upper half.

(c) Capital Market line (CML).

from our notes:

CML is the line from R_f tangent to the efficient frontier.

& its slope is called Sharpe Ratio.

So, our line starts at $R_f \rightarrow$ risk free portfolio & touches at efficient frontier with slope equals Sharpe Ratio.

④

(a) Standard error of the sample mean is:

$$SE(\hat{\mu}_i) = \frac{\sigma_i}{\sqrt{n}}$$

Given

$$\sigma_i = 30\%$$

$$n = 36$$

$$SE(\hat{\mu}_i) = \frac{30\%}{\sqrt{36}} = \frac{30}{6} = 5\%$$

$\hookrightarrow$ This implies historical estimates of returns can contain large uncertainty.

$\therefore$ relying only on historical data may lead to inaccurate portfolio decisions.

(b) Error maximisation: - $\hat{\alpha}$ can produce large changes in optimal portfolio weights in MVO.

Replacing historical estimates with ML-predicted returns helps reduce this problem.

The optimiser tends to give very high weights to assets whose returns are slightly overestimated. Replacing historical estimates with ML-predicted returns help reduce this problem.

Becon;

ML models use more patterns, giving more stable & accurate return predictions. This leads to more reliable portfolio weights.

(c) for N assets, covariance matrix contains $\frac{N(N+1)}{2}$ unique entries.

for eg $N=50$, then $\frac{50(51)}{2} = 1275$.

$\therefore$ covariance matrix contains 1275 unique parameters.